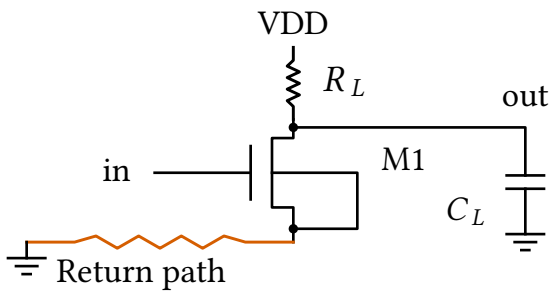
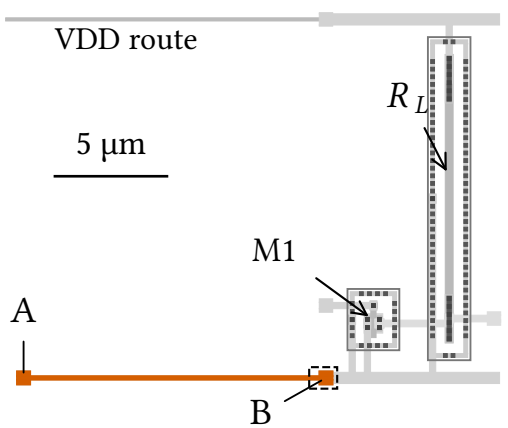
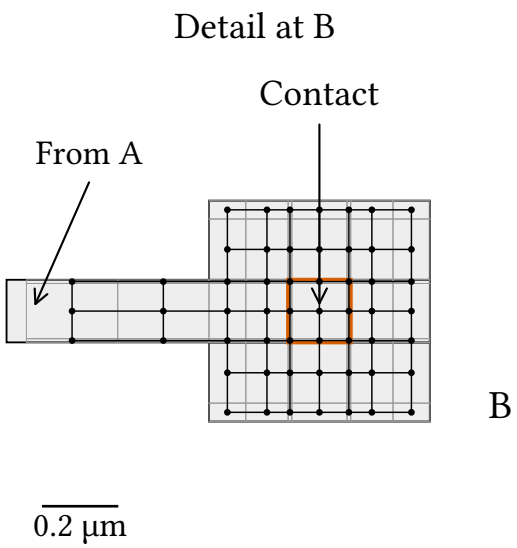
1. Circuit and layout





A: pad end · B: device end

2. Resistance network



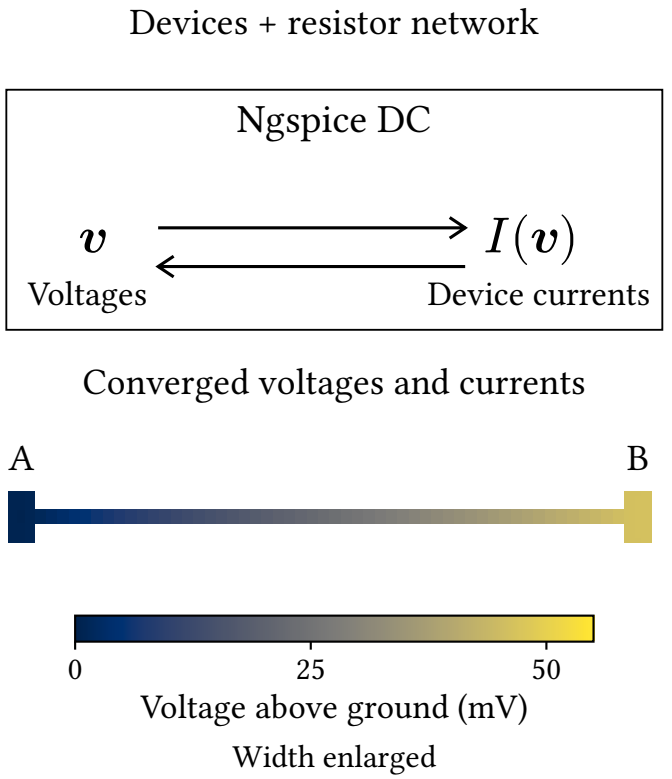
$$R_e = R_\square \frac{d_e}{w_e}$$

(edge resistance)

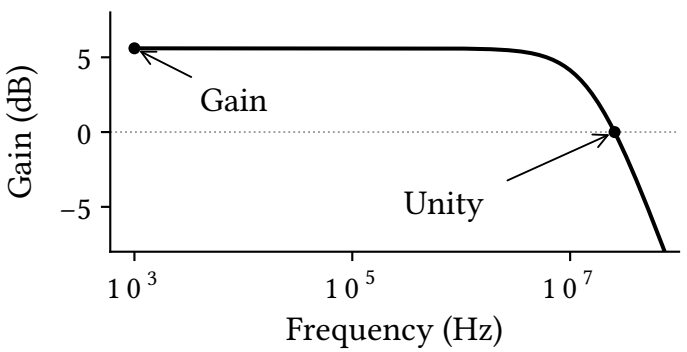
Wire and contact resistors

Solved circuit ⟶ metric response

3. Connected solution



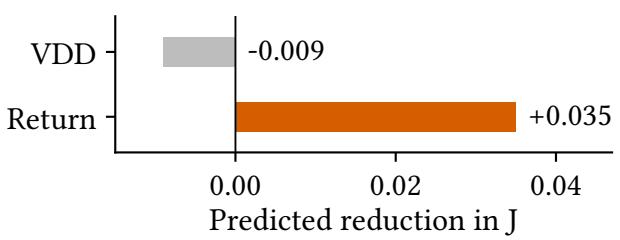
4. Objective sensitivity



Gain, f_u ⟶ J (target error)

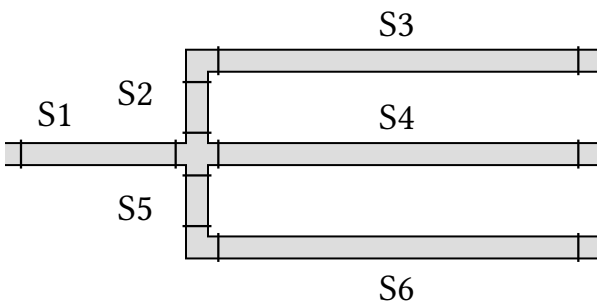
Adjoints ⟶ $\frac{dJ}{dR_e}$

1% span-resistance reduction



5. Automatic widening

Permitted routing ⟶ find spans



Every segment × permitted widths k

Segment	Benefit	Area	Per area
S1	$\hat{B}_{1,k}$	$\Delta A_{1,k}$	$\hat{\eta}_{1,k}$
S2	$\hat{B}_{2,k}$	$\Delta A_{2,k}$	$\hat{\eta}_{2,k}$
S3	$\hat{B}_{3,k}$	$\Delta A_{3,k}$	$\hat{\eta}_{3,k}$
S4	$\hat{B}_{4,k}$	$\Delta A_{4,k}$	$\hat{\eta}_{4,k}$
S5	$\hat{B}_{5,k}$	$\Delta A_{5,k}$	$\hat{\eta}_{5,k}$
S6	$\hat{B}_{6,k}$	$\Delta A_{6,k}$	$\hat{\eta}_{6,k}$

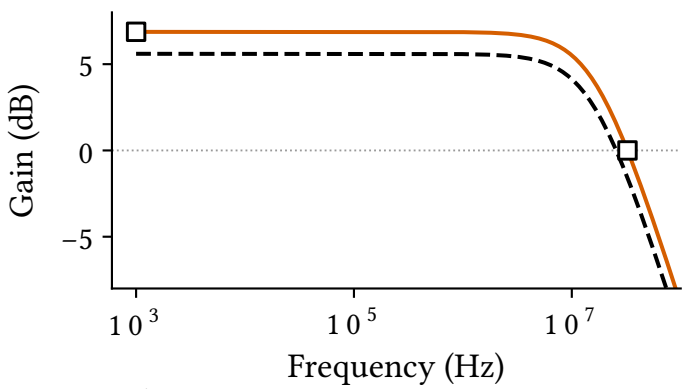
Metric limits + $\Delta A \le A_{\text{budget}}$

Refine ⟶ rank by total benefit

6. Verify ranked candidates

Fixed-span reference · 10 µm²

Saved predictions ⟶ full ngspice



Benefit: 1.980 predicted / 1.978 measured

Gain +15.76% (V/V) · f_u +26.77%

Select by measured benefit